

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 06 — INTRODUCTION TO MODBUS

Student Edition

Authorized ICDFA training environments only
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Lab 06: Introduction to Modbus

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 3–4 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Explain the difference between Modbus TCP and Modbus RTU.
- Identify common Modbus function codes and data areas.
- Observe Modbus TCP communication between FUXA and OpenPLC in an isolated lab.
- Use Wireshark to capture/filter Modbus traffic and identify request/response fields.
- Relate register/address information to the running tank-fill process.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	plc	password
SCADA VM	192.168.2.10	scada	password
Analyst/Kali VM	Instructor-provided	kali	kali

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Reference: Modbus Function Codes

Code	Function	Purpose
01	Read Coils	Read discrete output/coils
02	Read Discrete Inputs	Read discrete input states
03	Read Holding Registers	Read 16-bit holding registers
04	Read Input Registers	Read 16-bit input registers
05	Write Single Coil	Change one discrete coil
06	Write Single Holding Register	Change one holding register
15	Write Multiple Coils	Change multiple coils
16	Write Multiple Holding Registers	Change multiple registers

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Start the PLC, SCADA and analyst VMs on the instructor-provided isolated laboratory topology. Do not connect this lab to production or bridged networks.

Step 2. On the PLC VM, start the TankFillSim program in OpenPLC Runtime and confirm Running status.

Step 3. On the SCADA VM, start FUXA, load the tank-fill HMI and confirm the openplc connection targets 192.168.1.10:502.

Step 4. Run the HMI and verify that the tank level changes between its operating limits.

Step 5. On the analyst VM, open Wireshark. Select only the interface connected to the ICDFA lab segment carrying PLC/SCADA traffic.

Step 6. Apply the display filter modbus. If needed, refine it with `tcp.port == 502`.

Step 7. Start a short packet capture while the HMI is running, allow several process updates to occur, then stop the capture.

Step 8. Select a Modbus request/response pair and expand the Modbus/TCP and Modbus protocol fields.

Step 9. Identify the transaction/request, source and destination IP addresses, function code and referenced register/data value.

Step 10. Locate a Function Code 4 (Read Input Registers) response associated with the tank-level value. In the teaching image, Input Register 80 corresponds to the Tank Level indication.

Step 11. Complete the Modbus function-code table by explaining at least codes 01, 02, 03, 04, 05, 06, 15 and 16.

Step 12. Save the packet capture with a filename containing your registration number and LAB06. Do not capture unrelated traffic.

Required Evidence & Submission

- Screenshot of the FUXA/OpenPLC process running.
- Screenshot of Wireshark filtered to Modbus traffic.
- One annotated request/response analysis identifying function code and register.
- The saved .pcap/.pcapng capture from the isolated training network.
- A completed function-code table and a short comparison of Modbus TCP versus Modbus RTU.

Knowledge Check

1. What transport/networking difference separates Modbus TCP from Modbus RTU?

Response: Modbus TCP operates over Ethernet networks using the TCP/IP protocol stack (typically port 502), while Modbus RTU operates over serial communication links such as RS-232 or RS-485 and uses a binary frame structure with CRC error checking.

2. Why does Modbus use function codes?

Response: Modbus uses function codes to specify the action that a client (master) wants a server (slave) device to perform, such as reading coils, reading registers, or writing values to registers. The function code tells the receiving device how to process the request.

3. What does Function Code 4 request/read?

Response: Function Code 04 (Read Input Registers) is used to read the contents of one or more input registers from a Modbus device. In this lab, Input Register 80 contains the Tank Level value.

4. Why is visibility of unprotected industrial protocol traffic a cybersecurity concern?

Response: Many industrial protocols, including Modbus TCP, do not provide built-in encryption or authentication. An attacker who can access the network may view process data, monitor operations, intercept communications, or potentially manipulate control commands, creating risks to system safety, availability, and integrity.